

Eric Martin

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Summary

Senior Data Engineer with 9+ years of experience architecting and implementing scalable **AWS Data Lake architectures, Snowflake ELT workflows, and real-time streaming ingestion systems** within financial services environments. Strong hands-on expertise in building high-volume **ETL/ELT pipelines using S3, Glue, EMR, Kinesis, Lambda, PySpark, and SQL**, supporting transactional workloads and ML-ready feature datasets. Experienced in **Change Data Capture (CDC), incremental ingestion, partition optimization, performance tuning, dimensional modeling, CI/CD automation, and secure data governance frameworks** aligned with regulatory standards.

Technical Expertise

Cloud Platforms

Amazon Web Services (AWS), S3, Glue, EMR, Kinesis Data Streams, Lambda, IAM, Step Functions, CloudWatch, EC2, VPC

Microsoft Azure (ADLS Gen2, Azure Data Factory – integration exposure)

Data Engineering

ETL, ELT, Data Pipelines, Data Lake Architecture, Data Ingestion, Change Data Capture (CDC), Incremental Loading, Schema Evolution, Batch Processing, Streaming Processing, Data Partitioning, Performance Tuning, Data Migration, Data Integration, Data Modeling

Data Warehousing

Snowflake, Virtual Warehouses, Clustering Keys, Query Optimization, Micro-partitioning, Warehouse Scaling, RBAC, Data Sharing, Data Marts

Programming & Processing

Python, PySpark, SQL, Spark SQL, Spark Structured Streaming, Pandas, Shell Scripting

Streaming & Event Systems

Kinesis, Event-Driven Architecture, Real-Time Processing, Stream Processing, Message Queues, Event Ingestion Patterns

DevOps & Infrastructure

CI/CD, GitHub Actions, Git, Docker, Terraform, Infrastructure as Code (IaC), Deployment Automation, Monitoring & Alerting, Logging

Analytics & ML Support

Dimensional Modeling (Star Schema), Feature Engineering, Batch Inference Pipelines, Data Quality Checks

Security & Governance

IAM Policies, Role-Based Access Control (RBAC), Data Encryption, Audit Logging, Access Control, Compliance Alignment

Professional Experience

HSA Bank | Senior Data Engineer

January 2022 – Present

- Architected and implemented enterprise-scale **AWS Data Lake architecture** using S3 (raw, staged, curated zones), Glue jobs, and EMR Spark clusters to process over **80M+ financial transaction and account records monthly**, enforcing encryption, IAM policies, and access control frameworks.
- Designed and developed scalable **ETL and ELT pipelines** orchestrated via Glue and Step Functions, transforming raw transactional datasets into curated Snowflake schemas using incremental and CDC ingestion logic.
- Implemented **Change Data Capture (CDC) frameworks** using timestamp- and log-based extraction methods, reducing full table reloads and improving synchronization with downstream analytical systems.
- Built and maintained **near real-time streaming ingestion pipelines** leveraging Kinesis Data Streams and Lambda consumers, implementing checkpointing and failure recovery mechanisms for reliable event processing.
- Reduced batch processing runtime from approximately **5 hours to 2.8 hours** through Spark executor tuning, partition pruning strategies, and EMR cluster right-sizing.
- Developed optimized **Snowflake ELT workflows**, creating staging layers, clustering keys, and warehouse scaling policies to improve query performance and cost efficiency.
- Engineered curated **dimensional data models (fact and dimension tables)** supporting finance, compliance, and risk reporting dashboards.
- Designed idempotent ingestion and controlled backfill frameworks enabling safe historical data reprocessing without duplication.
- Built ML-ready feature engineering datasets supporting fraud detection and customer risk scoring batch inference pipelines.
- Implemented centralized monitoring using CloudWatch metrics, structured logging, and alert notifications improving SLA compliance.
- Partnered with governance teams to enforce RBAC, encryption standards, audit logging, and regulatory data retention policies.
- Automated deployments using **CI/CD pipelines (GitHub Actions)** integrating code validation, testing, and controlled environment promotion.
- Participated in cross-cloud integration efforts ingesting partner datasets originating from Azure-based pipelines into AWS environments.

Capital One | Senior Data Engineer

July 2017 – January 2022

- Designed and maintained distributed **AWS data pipelines** using S3, Glue, EMR, and Kinesis supporting high-volume credit card transaction and customer activity datasets.
- Developed **real-time streaming ingestion systems** using Kinesis and Spark Structured Streaming, implementing watermarking and checkpoint strategies to handle late-arriving events.
- Built scalable **Data Lake storage architecture** implementing partitioning strategies (date-based and hash-based) improving performance for multi-terabyte datasets.

- Implemented enterprise-grade **ELT frameworks loading curated data into Snowflake virtual warehouses** enabling fraud analytics and risk modeling use cases.
- Improved ETL latency by approximately **40%** through Spark memory configuration tuning, optimized join strategies, and distributed workload parallelization.
- Designed and maintained CDC ingestion pipelines synchronizing transactional systems with downstream analytical environments.
- Engineered curated **data marts and aggregated fact tables** supporting customer segmentation, credit risk scoring, and compliance reporting.
- Developed feature engineering workflows preparing ML-ready datasets for fraud detection and predictive analytics teams.
- Optimized Snowflake query performance via clustering key refinement, warehouse sizing adjustments, and micro-partition awareness.
- Implemented Infrastructure-as-Code practices using Terraform for reproducible AWS resource provisioning.
- Strengthened IAM role configuration, encryption standards, and secure key management aligning with financial compliance requirements.
- Contributed to CI/CD process improvements including automated testing and version-controlled ETL pipeline deployments.

Glorium Technologies | ETL Developer

July 2015 – June 2017

- Developed backend data processing services using Python and SQL supporting financial analytics and reporting systems.
- Assisted migration of legacy ETL workflows to AWS cloud infrastructure, participating in S3 storage design and Glue job development.
- Built ingestion utilities integrating REST APIs, CSV/JSON batch files, and relational databases into centralized repositories.
- Designed normalized relational schemas and optimized SQL queries supporting reporting workloads.
- Implemented logging, exception handling, and monitoring scripts improving batch job reliability.
- Supported DevOps initiatives including Docker containerization and Git-based release management.

Education

M.S. Data Science, Rowan University

May 2015 | Newark, NJ

B.S. Information Technology, Rowan University

May 2013 | Glassboro, NJ